

ZeroDrift MK2: the complete build

Every part, every wire, every command, from an empty table to the rig locking onto the beacon on the website. Follow the sections in order. Every step ends with a **CHECK**. Don't go on until the check passes.

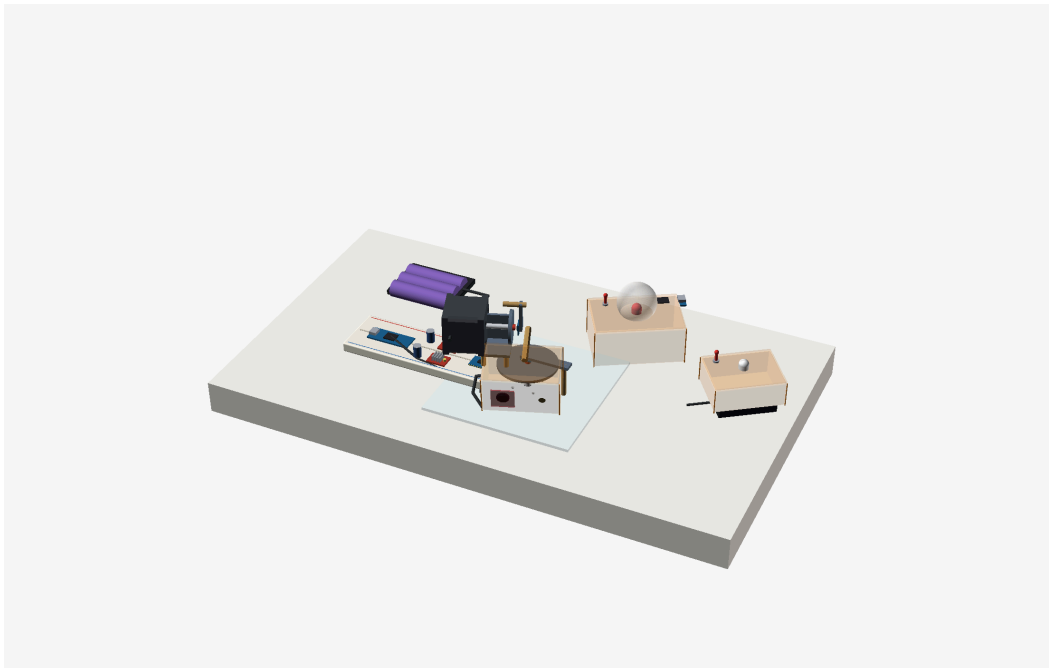
$\approx 7\text{ h}$
mechanics 3 h · wiring 2 h · bring-up 2 h

$0.11^\circ / \text{step}$
1/16 microstep, direct drive

0.088°
AS5600 encoder reading

$\pm 90^\circ / \pm 18^\circ$
pan / tilt travel limit

Tonight's order (direct drive, no belt). **1.** Shop: shopping list (≈ 30 min). **2.** Beacon + decoy, section 9 (≈ 1 h). Test the beacon on the Live page with the laptop camera right away. **3.** Rig mechanics, section 4 ($\approx 2\text{--}3$ h). **4.** Wiring, sections 5–6 (≈ 1.5 h). **5.** Firmware and first power-on, sections 7–8 (≈ 45 min). **6.** Website, calibration and demo, sections 10–11 (≈ 45 min). **7.** Record the demo script in 10.9. Also record the screen, so the Live page and the rig are both in the video.



What you are building: the MK2 gimbal (camera and laser in one head) tracking the 4 Hz red beacon, while ignoring the decoy: a red light just as bright that never blinks.

01 Parts: have / arriving / buy	02 Tools and software	03 Five safety rules
04 Mechanical build	05 Wiring: diagrams	06 Wiring: wire-by-wire list
07 Arduino: flash the firmware	08 First power-on and tests	09 Beacon (9A) and decoy (9B)
10 Connect to the website	11 Calibrate	12 Troubleshooting

01 Parts

HAVE in hand **ARRIVING** ordered **BUY** get it at Amar Robotics (see the shopping list PDF)

The rig

PART	QTY	STATUS	CHECK BEFORE YOU USE IT
☐ NEMA17 stepper, 1.8°/step	2	HAVE	4 wires each
☐ A4988 driver module (+ heatsinks)	2	HAVE	Small resistors near the chip say R100 , R050 or R200 . Note it (for Vref, step 8.2). Stick the heatsink on the chip.
☐ NEMA17 L-bracket	2	HAVE	Only the tilt motor needs one; the second is a spare
☐ 100 µF 25 V capacitor	2	HAVE	Striped leg is –
☐ AS5600 magnetic encoder	2	HAVE	Solder the header pins if loose
☐ Diametric magnet 6 × 2.5 mm	2	HAVE	Must be diametric (test in step 4.4)
☐ TCA9548A I ² C multiplexer	1	HAVE	Solder the header pins
☐ USB webcam (UVC)	1	ARRIVING	Plug into the laptop and open it on the Live page before you strip the housing
☐ KY-008 laser module	1	HAVE	Pins marked S , middle, –
☐ ABS project box ≈ 70 × 50 × 30 mm (camera head)	1	BUY	Webcam board and laser inside, two holes in the front
☐ Red filter: red transparent acrylic 2 mm, red cellophane or an old sunglasses lens	1	BUY	Hold it over a phone camera: the room goes dim, a red LED stays bright
☐ Arduino Nano (rig)	1	HAVE	Have the right USB cable (Mini-USB or USB-C)
☐ 12 V motor supply : 3×18650 pack in its holder or a 12 V 2 A adapter	1	HAVE / BUY	Section 5 (Power). Each 18650 must read 3.7–4.2 V.
☐ DC jack with screw terminals (if you use the 12 V adapter)	1	BUY	Matches the adapter plug (5.5 × 2.1 mm)
☐ SPST toggle switch (motor kill switch)	1	HAVE	
☐ 830-point breadboard + Dupont jumpers (M-M, M-F, F-F)	1	HAVE	
☐ White sheet (base plate, ≈ 150 × 150 mm)	1	HAVE	
☐ Clear acrylic 3 mm, A4 + acrylic cutter (hook blade)	1	BUY	Pan disc 60 mm, tilt disc 40 mm, 2 encoder arms 15 × 60 mm, head plate. Score 5–6 times along a ruler, then snap.
☐ M3 × 10 screws, nuts, washers	20 / 30 / 20	BUY	Motors (M3 only), couplings to discs, L-bracket
☐ M3 × 40 screws + nuts	4 + 12	BUY	Posts for the two encoder arms
☐ M3 standoff 15 mm	4	BUY	Risers under the L-bracket

☐	M3 brass standoff 20 mm	4	BUY	Two screwed together = 40 mm head post. Brass, not plastic.
☐	M4 × 60 bolts + nuts	4 + 12	BUY	Base legs, so the pan motor hangs free
☐	5 mm rigid flange coupling	2	BUY	Bore 5 mm , rigid metal, grub screw. Holds the pan disc and the tilt disc on the shafts with no play.
☐	USB-C hub	1	BUY only if needed	Only if the laptop has USB-C ports only: the Nano and the webcam must be plugged in together

The beacon (section 9A)

PART	QTY	STATUS	CHECK
☐ Arduino Nano (second one) + USB cable	1	HAVE	
☐ Red LED, bright, 10 mm (5 mm works)	1	HAVE	Long leg = + (anode)
☐ 220 Ω resistor (red-red-brown)	1	BUY	150–330 Ω also works
☐ Ping-pong ball (half of it)	1	HAVE	Diffuser: makes the LED visible from the side
☐ Power: USB power bank, or your 5 V 1 A adapter	1	HAVE	Section 9A.6
☐ ABS project box ≈ 100 × 60 × 25 mm	1	BUY	LED goes through the lid

The decoy (section 9B)

PART	QTY	STATUS	CHECK
☐ 3×AA battery holder + 3 AA cells	1	HAVE	Fresh cells: ≈ 4.5–4.8 V together
☐ Red LED , same kind as the beacon	1	BUY	The decoy must look as bright as the beacon through the red filter . A white LED looks dim there.
☐ 100 Ω resistor (brown-black-brown)	1	BUY	82–150 Ω also works; two 220 Ω side by side = 110 Ω
☐ Toggle switch	1	HAVE	
☐ Glue and tape: Fevikwik, hot glue sticks, foam tape, insulation tape, zip ties	1	BUY	Shopping list items 17–21
☐ Other half of the ping-pong ball	1	HAVE	Same diffuser as the beacon: the only difference left is the blink
☐ ABS project box ≈ 100 × 60 × 25 mm	1	BUY	

02 Tools and software

Tools

☐ Multimeter : Vref, battery voltage, finding motor coil pairs. You can't build this without one.
☐ Soldering iron + solder: header pins on the TCA9548A and AS5600s

Software

☐ Arduino IDE 2 (arduino.cc/en/software)
☐ Library AccelStepper by Mike McCauley (Library Manager). Wire is built in.

- ☐ Small screwdriver (plastic/ceramic ideal) for the A4988 pot; Phillips for M3
- ☐ Drill + 3 mm and 5–6 mm bits (a step bit if you have one); masking tape for acrylic
- ☐ Hacksaw / acrylic cutter, file or sandpaper
- ☐ Allen key for the flange-coupling grub screws
- ☐ Superglue (magnets), hot glue (camera, laser), double-sided foam tape, zip ties, insulation tape
- ☐ Wire stripper; ruler; marker; a printed protractor or phone level app
- ☐ Two old credit cards: stacked, they make a 1.5 mm gap gauge for the encoders

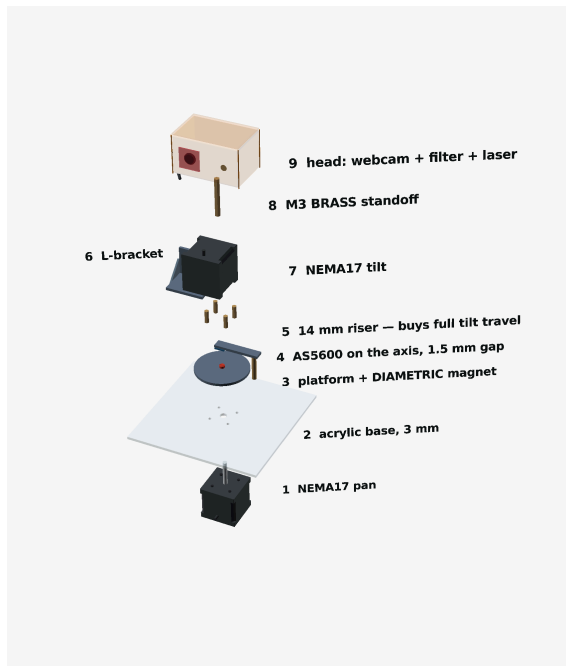
- ☐ Firmware from the repo:
`tools/rig/rig_firmware_v2.ino` (rig) and
`tools/rig/beacon_firmware.ino` (beacon)
- ☐ **Chrome or Edge** (the website talks to the Nano over Web Serial; Safari/Firefox can't)
- ☐ Clone Nano (CH340 chip) not showing a port: install the CH340 driver

03 Five safety rules (each one kills a part)

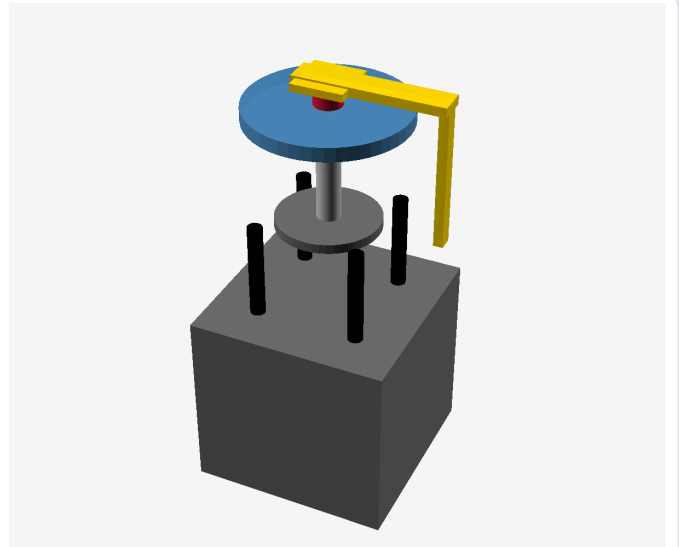
- 1 Never plug or unplug a motor while the 12 V switch is on.** The A4988 dies instantly and silently.
- 2 Capacitor on every VMOT,** 100 μ F, right at the driver pins, correct way round (stripe to GND). Without it the first power-on spike can kill the driver.
- 3 One common ground.** Battery –, both drivers, the Nano and everything else share GND. Missing ground = motors buzz or do nothing.
- 4 Laser:** never look into it and never point it at people. Test on a wall.
- 5 12 V never goes to the Nano, the 5 V rail, VDD, the TCA or the AS5600s.** 12 V only touches VMOT. Check the rails with the multimeter before plugging boards in.

04 Mechanical build

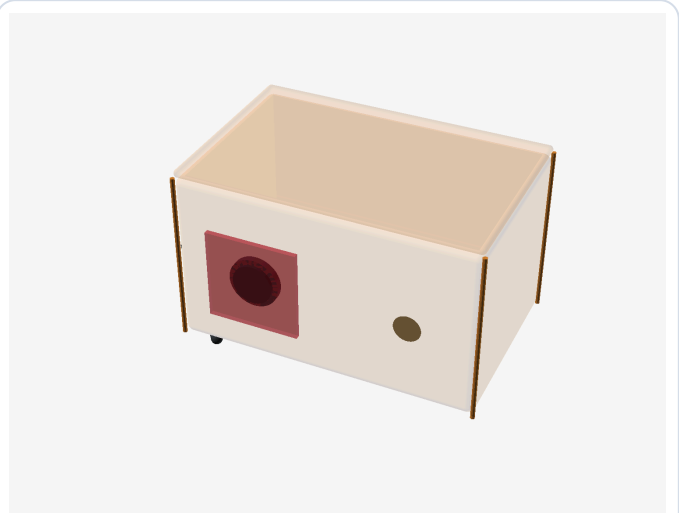
The 3D step-by-step assembly shows every part below arriving in its place: click a stage, orbit, explode. Keep it open next to you.



Bottom to top: 1 pan motor · 2 base · 3 platform + magnet · 4 pan AS5600 · 5 risers · 6 L-bracket · 7 tilt motor · 8 brass standoff · 9 head.



The AS5600 chip sits exactly on the axis, 1.5 mm from a diametric magnet on the part that turns. The arm holding the chip must NOT turn.



Head: camera behind the red filter, laser beside it, both looking the same way.

- 4.1 Base plate (white sheet).** Keep the protective film on, and put masking tape where you'll drill. Mark the centre. Hold the pan motor under the sheet, face up, and mark through its 4 screw holes (a 31 mm square). Drill the 4 holes at 3 mm and the centre at 6 mm. Drill slowly, no pressure, or acrylic cracks.

CHECK The motor shaft passes through the centre hole without touching the edge.

2 4.2 Pan motor. Bolt the motor **under** the base, shaft pointing up, with 4 M3×10 screws. Put 2 nuts on each screw between the motor and the plate as spacers: the motor's round boss (22 mm, 2 mm tall) must not press on the acrylic. Put an **M4 × 60 bolt** through each corner as a leg (a nut above and below the plate, and a nut at the bottom to level it), so the motor hangs free. Put the motor cable out to the side you'll wire from.

CHECK The base sits flat and doesn't rock; the motor doesn't touch the table.

3 4.3 Pan platform + magnet. Cut a 60 mm disc from the clear acrylic (score around a drawn circle, or cut a hexagon and file the corners). Draw 2 lines through its centre (a cross). Bolt it on top of a **5 mm flange coupling**, centred. Slide the coupling onto the pan shaft and leave a 2 mm gap to the base (the credit cards make a good spacer). Tighten the grub screw onto the flat of the shaft. Superglue a **diametric** magnet exactly on the centre of the cross.

CHECK Turn the motor shaft by hand: the disc spins flat, with no wobble and no rubbing. The magnet stays on the centre and doesn't draw a circle. If the disc wobbles, re-seat the coupling; last resort is the lazy-Susan bearing.

4 4.4 Magnet test (2 minutes, saves hours). Hold the second magnet near the glued one. A **diametric** magnet has its N and S on the *sides*: rolling one around the other flips attract/repel as you go round the edge. An axial magnet has N/S on the flat faces; it will read a constant and the encoder is dead with no error.

CHECK Both magnets are diametric.

5 4.5 Pan encoder arm (does NOT turn). Stand an **M3 × 40 screw** up through the base about 38 mm from the centre (nut underneath, nut on top). Cut a clear acrylic strip ≈ 15 × 60 mm, drill a 3 mm hole at one end, and fix it on top of the screw between two nuts so it reaches over the centre. Fix the pan AS5600 under the arm, **chip face-down, chip exactly over the magnet**. Set the height with the 2 credit cards between chip and magnet (1.5 mm; 0.5–3 mm works). Use washers/nuts on the post to adjust.

CHECK Looking straight down, the chip is centred on the magnet within 0.5 mm, and the arm stays still when the platform turns.

6 4.6 Risers + L-bracket. Four **15 mm M3 standoffs** on the platform, L-bracket on top of them. The risers lift the bracket over the encoder arm. Check this in the 3D view. Put the bracket slightly off-centre so the tilt shaft ends up near the pan axis.

CHECK Turn the platform ±90° by hand: the bracket and risers pass over and around the encoder arm and post without touching.

7 4.7 Tilt motor. Bolt it to the L-bracket upright, shaft **horizontal**, at right angles to the way the head looks, so tilt nods up and down.

CHECK The shaft is level (phone level app).

8 4.8 Tilt disc + magnet + encoder. A 40 mm acrylic disc on the second flange coupling, on the tilt shaft. Glue the second diametric magnet at the centre of the disc's outer face. Fix the tilt AS5600 on a second clear acrylic strip (with an M3 × 40 post if needed) bolted to the L-bracket (the side that does NOT turn with tilt). The chip faces the magnet, centred on the shaft line, 1.5 mm away.

CHECK Nodding the disc by hand, the gap stays at 1.5 mm and nothing rubs.

9 4.9 Brass standoff + head. Screw two **20 mm brass** M3 standoffs together (≈ 40 mm). Bolt this to the tilt disc near its edge and hang the head box (the $70 \times 50 \times 30$ mm ABS box) from it. Never use a printed or plastic part here: it sags and shakes. Inside the box, take the webcam board out of its housing. Hot-glue it behind a hole in the front, with the **red filter** over the lens (filter over the camera only, not the laser). Hot-glue the KY-008 laser beside it, about 20 mm away, pointing the same way (parallel).

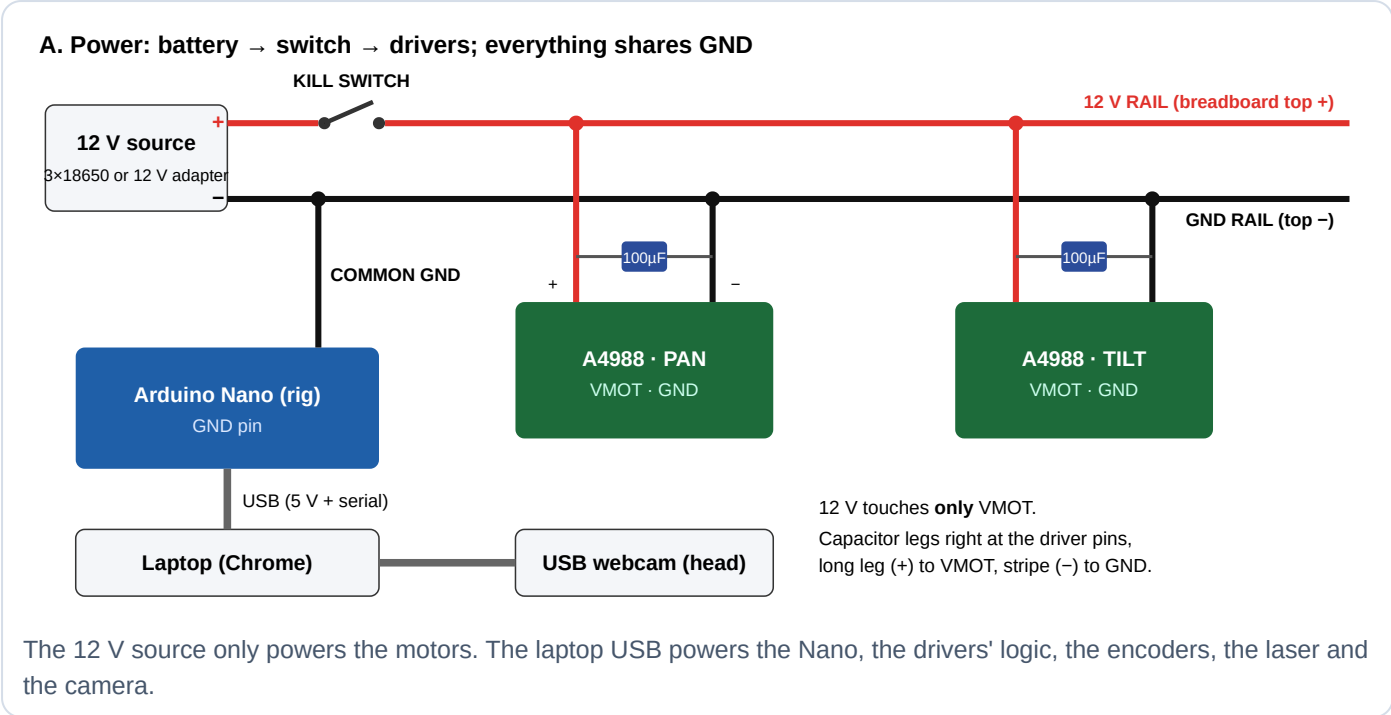
CHECK With the motors unpowered, move the head: it swings freely. Tilt $\pm 18^\circ$ clears the platform. Camera and laser point the same way (laser dot near the middle of the camera view on a wall 2 m away).

10 4.10 Cables. Run the camera USB, laser wires, tilt-motor cable and tilt-encoder wires down the bracket together (zip ties). Leave a loose **service loop** at the tilt joint and at the pan joint so full travel never pulls a wire. Keep the loop off the base plate.

CHECK Pan $-90^\circ \rightarrow +90^\circ$ and tilt $-18^\circ \rightarrow +18^\circ$ by hand: no cable goes tight or catches.

05 Wiring: diagrams

Wire colours used below: **red = 12 V**, **orange = 5 V**, **black = GND**, **blue = SDA**, **green = SCL**, **purple = signals**. Do all wiring with the battery **unplugged** and the Nano USB **unplugged**.



5.0 Power: what connects to what

The rig has **two separate power sources**, joined by one ground wire:

SOURCE	POWERS	HOW
Laptop USB (stays connected the whole time)	Nano, A4988 logic (VDD), TCA9548A, both AS5600s, laser, webcam	USB cable into the Nano; webcam USB into the laptop
12 V source: 3×18650 pack in its holder or a 12 V 2 A adapter	Only the two motors, through the A4988 VMOT pins	Through the kill switch to the 12 V rail (wires 1–3)
One ground wire	Joins both sources	12 V source – ↔ Nano GND (wires 4–5). Without it nothing moves.

Never connected to the rig: the **B3 balance charger** (it only charges batteries, from the wall, with nothing else attached), and your **5 V 1 A adapter** (too weak for the motors; the A4988 needs at least 8 V). The 5 V adapter can power the **beacon** instead (section 9A).

1

Using the 18650 pack? Before putting the cells in the holder, set the multimeter to DC V and measure each cell.

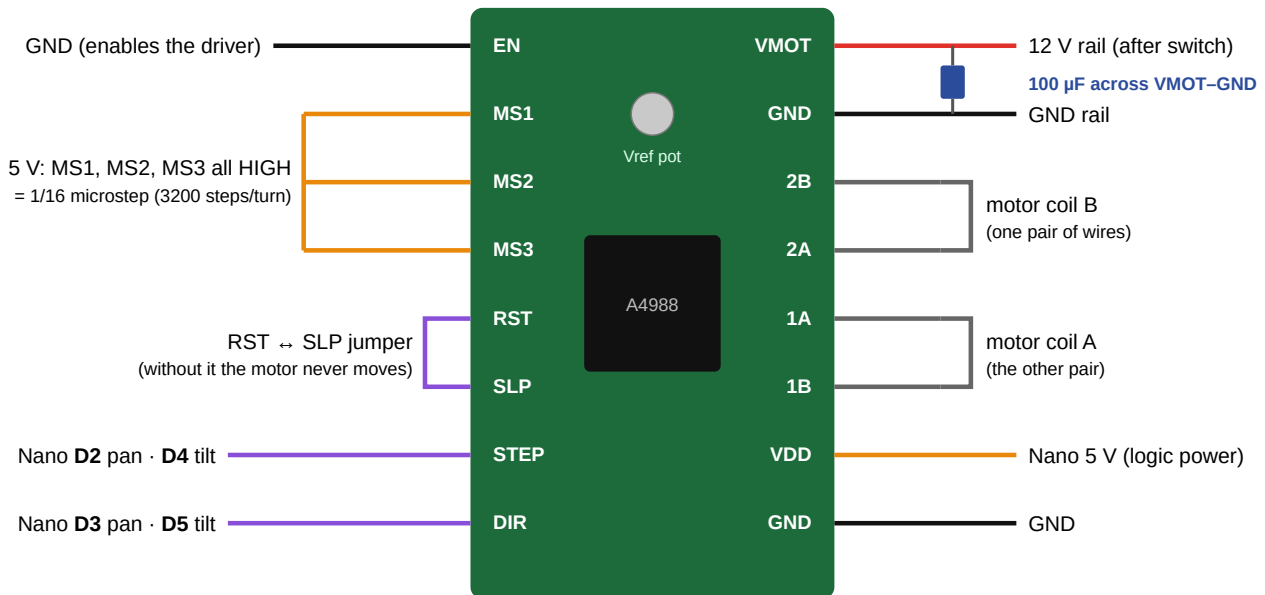
CHECK Each reads 3.7–4.2 V, and all three are within 0.1 V of each other. In the holder, the pack reads 11.1–12.6 V.

2 Cells below 3.5 V, or too far apart? Charge them **one at a time, out of the holder**, on a TP4056 module (the B3 can't: it needs a balance plug the holder doesn't have). Never charge the three in series. Or skip the battery and use a 12 V 2 A adapter with a DC jack.

3 Polarity. Holder: red = +, black = -. DC jack: the terminals are marked + and -. Mark the + wire with red tape before connecting anything.

CHECK Multimeter on the 12 V rail (switch ON, nothing else connected yet): +11 to +12.6 V. A minus sign means + and - are swapped.

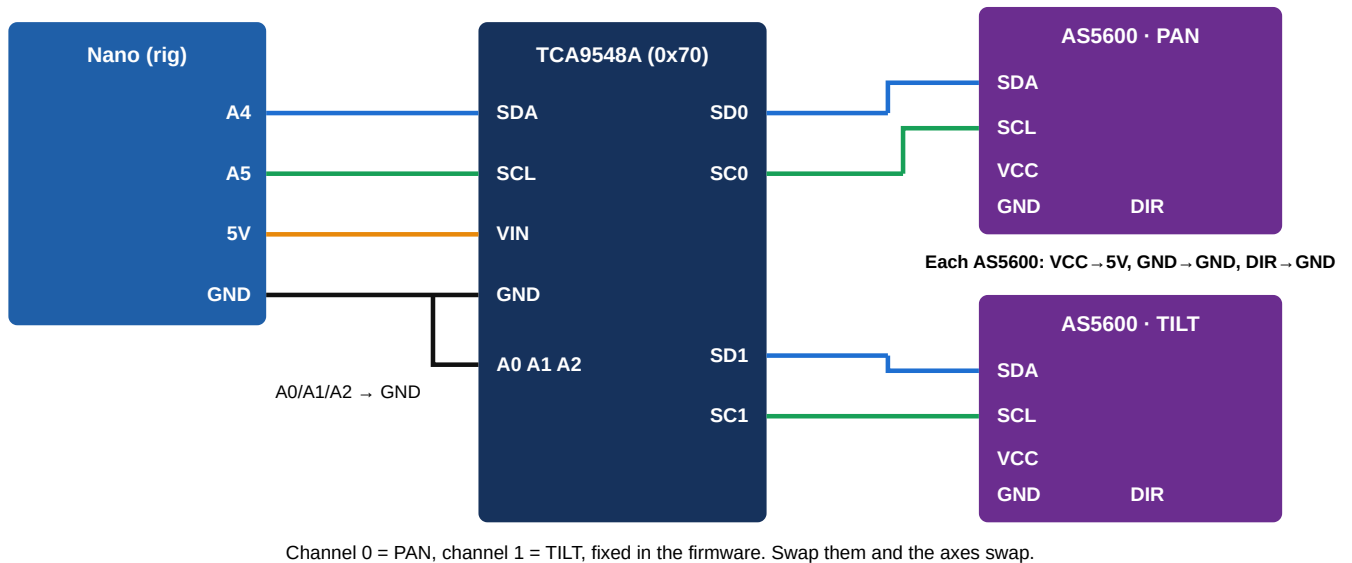
B. One A4988 driver, every pin (wire both the same way)



Pin order is for the common green/red A4988 board. Read the labels printed on YOUR board and follow the labels.

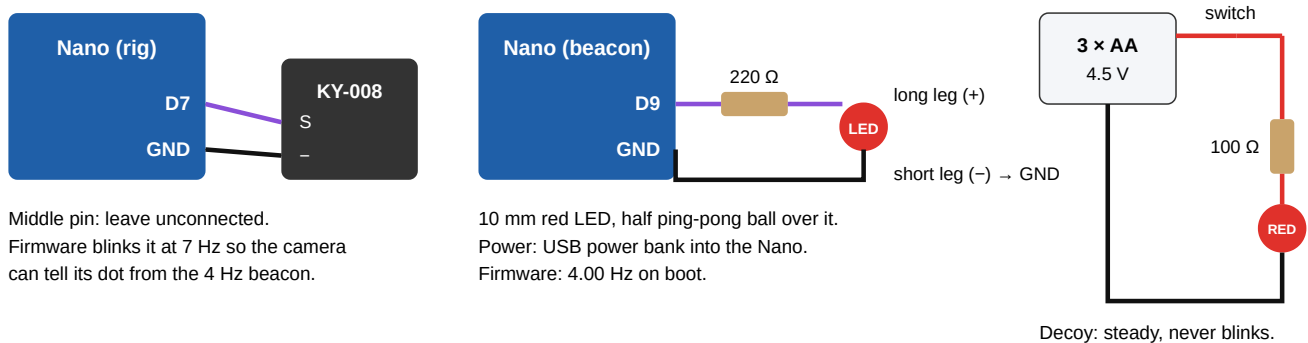
Find the coil pairs with the multimeter on continuity/ Ω : two wires that read $\approx 1\text{--}4\ \Omega$ are one coil. One pair goes to 1A/1B, the other to 2A/2B. If the motor turns the wrong way, the website can INVERT it later; don't rewire.

C. Encoders: Nano → TCA9548A → two AS5600 (both are address 0x36, so the mux is required)



Tie DIR to GND (a floating DIR makes the reading jump). If an AS5600 board has a closed "3V3" solder jumper, power that board from the Nano's 3V3 pin instead of 5 V. On the rig, the tilt encoder's 5 wires ride the pan axis: give them a service loop.

D. Laser (on the rig) · beacon (second Nano) · decoy (no Nano)



The beacon Nano can drive the LED straight from D9 through 220 Ω (≈ 14 mA). The 2N2222 in the kit is not needed.

06 Wiring: wire-by-wire list (tick each one)

Breadboard layout: top rail pair = **12 V / GND**, bottom rail pair = **5 V / GND**. Nano in the middle, the two A4988s either side, TCA9548A at one end. Leave the 12 V rail's + side empty until step 8.3.

#	FROM	TO	NOTE
Power and ground			
☐ 1	12 V source + (battery holder red / DC jack +)	Kill switch, pin 1	Red wire
☐ 2	Kill switch, pin 2	Top rail +, the 12 V rail	
☐ 3	12 V source – (battery holder black / DC jack –)	Top rail –, the GND rail	
☐ 4	Top GND rail	Bottom GND rail	Common ground. Don't skip it.
☐ 5	Nano GND	Bottom GND rail	
☐ 6	Nano 5V	Bottom + rail, the 5 V rail	Never connect this to the 12 V rail
PAN driver (A4988 #1)			
☐ 7	12 V rail	PAN VMOT	
☐ 8	GND rail	PAN GND (next to VMOT)	
☐ 9	100 µF: long leg / stripe leg	PAN VMOT / PAN GND	Right at the pins
☐ 10	5 V rail	PAN VDD	
☐ 11	GND rail	PAN GND (next to VDD)	
☐ 12	5 V rail	PAN MS1, MS2, MS3	3 short jumpers → 1/16 step
☐ 13	PAN RST	PAN SLP	Short jumper between the two pins
☐ 14	PAN EN	GND rail	
☐ 15	Nano D2	PAN STEP	
☐ 16	Nano D3	PAN DIR	
☐ 17	Pan motor coil A / coil B	PAN 1A-1B / 2A-2B	Pairs from the multimeter test
TILT driver (A4988 #2): same as PAN, different Nano pins			
☐ 18–25	Repeat wires 7–14 on the TILT driver		VMOT, GND, cap, VDD, GND, MS1-3, RST-SLP, EN
☐ 26	Nano D4	TILT STEP	
☐ 27	Nano D5	TILT DIR	
☐ 28	Tilt motor coil A / coil B	TILT 1A-1B / 2A-2B	
Encoders			

☐	29	Nano A4	TCA9548A SDA	Blue
☐	30	Nano A5	TCA9548A SCL	Green
☐	31	5 V rail / GND rail	TCA9548A VIN / GND	
☐	32	GND rail	TCA9548A A0, A1, A2	Address 0x70
☐	33	TCA SD0 / SC0	PAN AS5600 SDA / SCL	Channel 0 = pan
☐	34	5 V rail / GND rail / GND rail	PAN AS5600 VCC / GND / DIR	
☐	35	TCA SD1 / SC1	TILT AS5600 SDA / SCL	Channel 1 = tilt
☐	36	5 V rail / GND rail / GND rail	TILT AS5600 VCC / GND / DIR	
Laser and USB				
☐	37	Nano D7	KY-008 S	
☐	38	GND rail	KY-008 –	Middle pin: nothing
☐	39	Nano USB	Laptop	Serial + 5 V
☐	40	Webcam USB	Laptop (hub if needed)	

Not used in this build: D8 (vibration injector) and D9/D10 (old servo fine stage). Leave them empty.

07 Arduino: flash the firmware

- 1 Arduino IDE → **Tools** → **Manage Libraries** → search `AccelStepper` → install (Mike McCauley).
- 2 Open `tools/rig/rig_firmware_v2.ino`. No edits needed: upload it as it is.
- 3 **Tools** → **Board** → **Arduino Nano. Processor** → **ATmega328P**. If upload fails with "not in sync", choose **ATmega328P (Old Bootloader)** (most clones).
- 4 **Tools** → **Port** → the Nano (appears when you plug it in; `usbserial` / `wchusbserial` on Mac, `COMx` on Windows). Click **Upload**.
- 5 **Tools** → **Serial Monitor**. Set **115200 baud** and line ending **Newline**.
CHECK It prints `# ZeroDrift Mk2 ready, encoders=yes`. "encoders=no" → section 12.

08 First power-on and tests

- 1 8.1 Rails check (battery unplugged, Nano on USB).** Multimeter DC V: 5 V rail to GND $\approx$ 4.7–5.0 V. The 12 V rail reads 0.

CHECK 5 V present; nothing is hot.

- 2 8.2 Set Vref on both drivers (Nano USB only, battery OFF).** Black probe on GND, red probe touching the metal of the driver's little pot. Turn gently to read:

RESISTOR MARKING	R100 (MOST COMMON)	R050	R200
Vref ($\approx$ 0.7 A per coil)	0.55 V	0.28 V	1.1 V

CHECK Both drivers set. (Vref = current $\times$ 8 $\times$ resistor. The motors hold firmly, and the drivers only get warm.)

- 3 8.3 Motors on.** Motors plugged in. Flip the kill switch ON.

CHECK Both shafts become stiff (holding). No smell, no hot chip after 1 minute. If a driver gets too hot to touch, lower Vref 10 %.

- 4 8.4 Serial tests.** Type each line into the Serial Monitor and press Enter:

you type	what should happen
?	S 0 0 E 0 E = encoder reading (C = no encoders)
P267 T0	pan turns 30° (8.889 steps per degree)
?	S 26x 0 E 0 pan $\approx$ 267 $\pm$ 3 from the encoder
P0 T0	pan returns
T89	tilt nods 10°
T0	tilt returns
L1	laser ON (steady)
L7.0	laser blinks 7 Hz
L0	laser OFF
!	# encoder reads 10.0x deg (expect ~10.00 ...)
Z	# zeroed here = straight ahead

CHECK Pan and tilt both move, both come back, and the encoder numbers match the command within a few steps.

- 5 8.5 Encoder direction.** After P267 T0, ? must say about +267. If it says -267, move that AS5600's DIR wire from GND to 5 V. Do the same for tilt with T89.

CHECK Encoder sign equals the command sign on both axes.

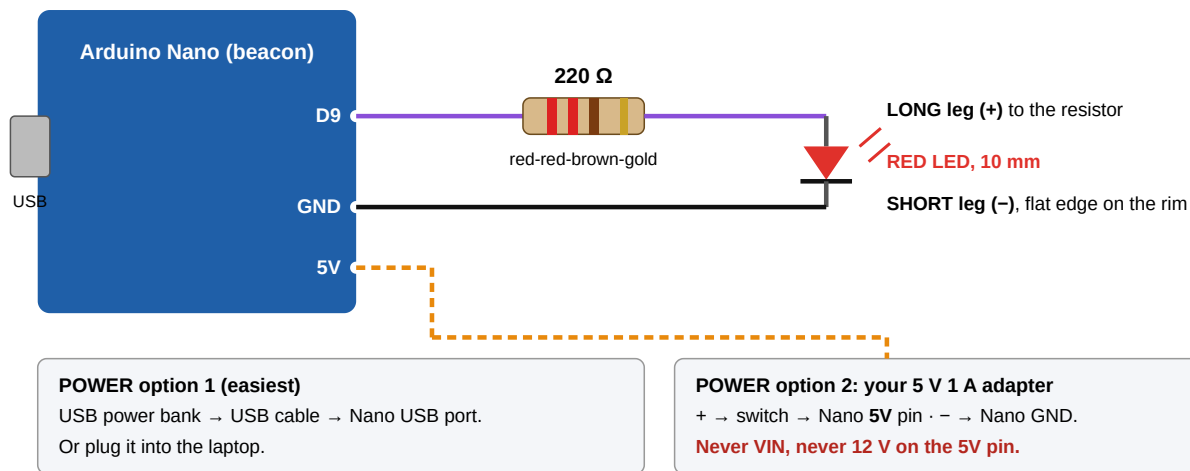
- 6 8.6 Close the Serial Monitor,** or the website can't open the port.

09 Beacon (9A) and decoy (9B)

These are the two targets. The **beacon** is the thing the rig must find and follow: a red LED blinking at exactly **4 Hz**. The **decoy** is the test: a red LED of the same colour, size and brightness that **never blinks**. A tracker that goes by brightness would lock onto the decoy. Ours only locks the 4 Hz blink. Build them to look the same, so the blink is the only difference.

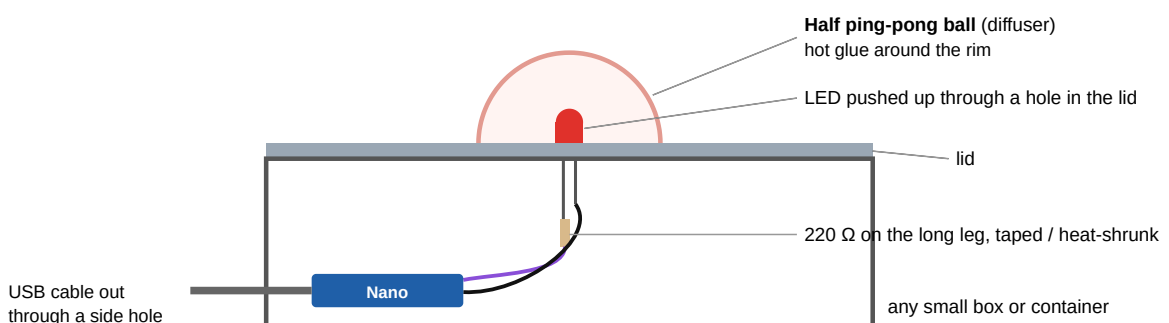
9A · The beacon

9A-1. Beacon circuit



Only three connections: D9 → 220 Ω → LED long leg; LED short leg → GND. The Nano blinks it at exactly 4.00 Hz. (150–330 Ω also works; 1 k Ω is too dim.)

9A-2. Beacon assembly (cut through the middle)



The LED sits in the centre of the dome. The ping-pong ball spreads its light, so the camera sees a bright ball from any angle, not just a dot when it faces the LED head-on.

- 9A.1 Flash the beacon firmware.** Plug the second Nano into the laptop. Arduino IDE → open `tools/rig/beacon_firmware.ino` → Board: Arduino Nano → Processor: ATmega328P (or Old Bootloader) → Port → **Upload**. Open the Serial Monitor at **115200**, line ending **Newline**.
CHECK It prints `ZeroDrift beacon ready F4.00 B255 M1` (4 Hz, full brightness, blinking).

- 2 **9A.2 Prepare the LED.** Find the **long leg (+)**. The short leg (–) is on the side with a flat edge on the LED's rim. Twist one end of the **220 Ω** tightly around the long leg and solder it (or twist and tape). Cover the joint with heat-shrink or tape so it can't touch the other leg.

CHECK The two legs can't touch each other.

- 3 **9A.3 Connect and test on the table.** A female–female jumper from the resistor's free end to Nano **D9**. Another from the LED's **short leg** to Nano **GND**.

CHECK The LED blinks steadily: on-off 4 times per second. No light at all = LED backwards; swap the two jumpers on the LED.

- 4 **9A.4 Test the commands** (Serial Monitor), one per line:

F6.0	OK F6.00	blinks 6 times a second
F4.0	OK F4.00	back to 4 Hz: the demo rate
B128	OK B128	half brightness (B255 = full)
M0	OK M0	steady on (no blink)
M1	OK M1	blinking again

CHECK Every command answers OK and the LED changes. After power-off it always starts again at 4 Hz, full brightness.

- 5 **9A.5 Box and diffuser.** Make a hole in the centre of the lid the size of the LED (10 mm or 5 mm): a drill, or a heated screwdriver tip on a plastic lid. Push the LED up through it from inside and hot-glue it underneath. Cut the ping-pong ball in half along its seam (cutter or small scissors). Keep the second half for the decoy. Hot-glue one half over the LED like a dome, LED in the centre. Put the Nano inside and run the USB cable out through a side hole.

CHECK Lid closed, the whole dome glows red on each blink.

- 6 **9A.6 Power. Option 1:** USB power bank into the Nano. Some power banks switch themselves off after about 30 s because the beacon draws so little. If yours does, use the laptop, a different power bank, or option 2. **Option 2:** your **5 V 1 A adapter**. If it has a USB socket or cable, plug it into the Nano. If it has bare wires or a barrel plug: + → toggle switch → Nano **5V** pin, – → Nano **GND**. Never use the VIN pin (it needs 7–12 V), and never put 12 V on the 5V pin. Unplug the adapter while the Nano is on the laptop's USB.

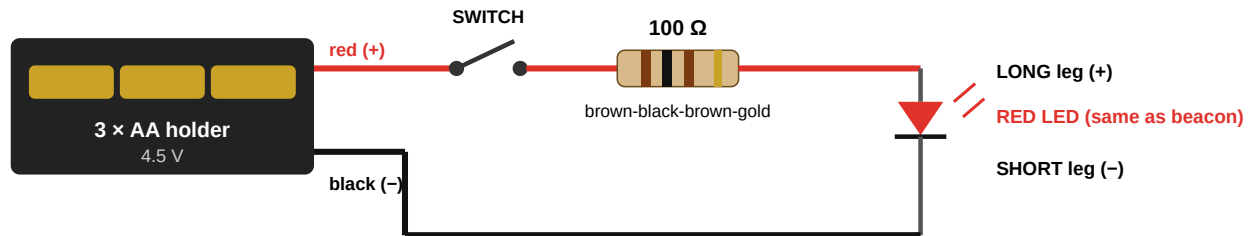
CHECK Left alone for 5 minutes, it is still blinking.

- 7 **9A.7 Rate check.** Film the beacon with your phone in slow motion for 2 seconds and count the flashes, or open the Live page on the laptop camera.

CHECK 4 flashes per second. The Live page locks it and shows about 4.0 Hz.

9B · The decoy

9B-1. Decoy circuit: no Arduino, no code



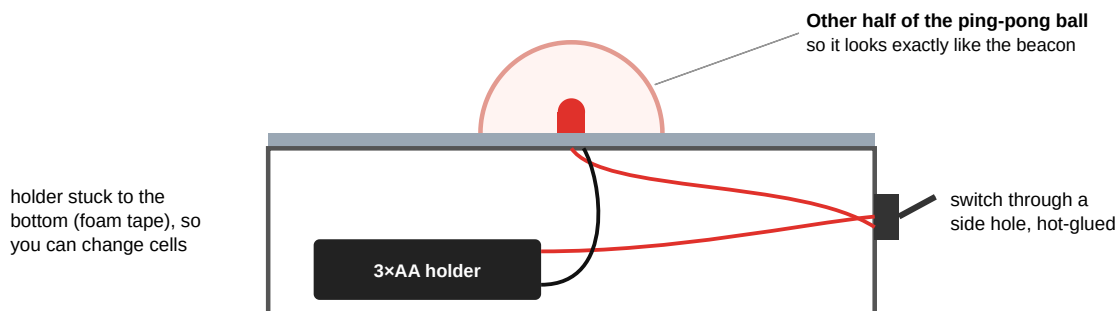
Why 100 Ω:

4.5 V from the cells – about 2 V across a red LED = 2.5 V across the resistor → $2.5 \div 100 = 25$ mA (bright, safe).

No 100 Ω? Two 220 Ω side by side (legs twisted together at both ends) = 110 Ω. Anything from 82 Ω to 150 Ω works. Never zero.

One loop: battery + → switch → 100 Ω → LED long leg → LED short leg → battery -. It is on and steady whenever the switch is on.

9B-2. Decoy assembly (cut through the middle)



Same box, same LED and same dome as the beacon. The only difference the camera can see is that the decoy never blinks.

- 1 **9B.1 LED + resistor.** Twist and solder the **100 Ω** onto the red LED's **long leg**. Heat-shrink or tape the joint.

CHECK The legs can't touch each other.

- 2 **9B.2 Switch.** Holder **red** wire → the switch's **middle** pin. One **outer** pin → the resistor's free end. Solder, or twist and tape.

- 3 **9B.3 Close the loop.** The LED's **short leg** → the holder's **black** wire.

- 4 **9B.4 Test.** Put 3 AA cells in (match the + and – marks in the holder) and flip the switch.

CHECK Steady red light, no blinking. No light: the LED is backwards (swap its legs), or the switch uses the other outer pin (flip it the other way).

5 **9B.5 Box.** A hole in the lid for the LED, the same as the beacon: LED through it, hot glue underneath, the **other half of the ping-pong ball** glued over it. A side hole for the switch, hot-glued. Stick the holder to the bottom with foam tape.

CHECK Next to the beacon, the two look the same, except that one blinks.

6 **9B.6 Brightness match** (on the Live page, with the MK2 camera and its red filter).

CHECK The decoy looks **as bright or brighter** than the beacon on screen. If it's clearly dimmer, use a smaller resistor (82 Ω or 68 Ω ; never below 56 Ω) or new cells.

7 **9B.7 The real test.** Beacon and decoy side by side, both on.

CHECK The rig locks the beacon every time and never the decoy, even with the decoy switched on after the lock.

Optional second decoy: a white LED with its own 100 Ω in the same box, wired in parallel with the red one (its own resistor, same switch). It looks bright to people in the room, but through the red filter it's weak. Show it only after the red decoy has passed.

10 Connect to the website

Open zerodrift-fsoc-pat.netlify.app/live in Chrome or Edge.

1 **START CAMERA**, then **MK2 CAMERA (USB)**. The view switches to the head's camera (HUD says *MK2 CAM*).

CHECK Wave in front of the head: you see it on screen.

2 **CONNECT RIG** → pick the Nano's port.

CHECK The RIG card says **MK2 · encoders**.

3 **MK2 DRIVE = Direct drive (1:1)**.

4 **TEST MOTION:** it should pan right, tilt up, then blink the laser.

CHECK Wrong direction → tick **INVERT PAN / INVERT TILT**. Wrong axis → **SWAP PAN / TILT**. Repeat until correct.

5 **BEACON RATE** = 4 Hz (the beacon's rate). Only that rate ($\pm 10\%$) is locked; a 3 Hz or 5 Hz light is ignored.

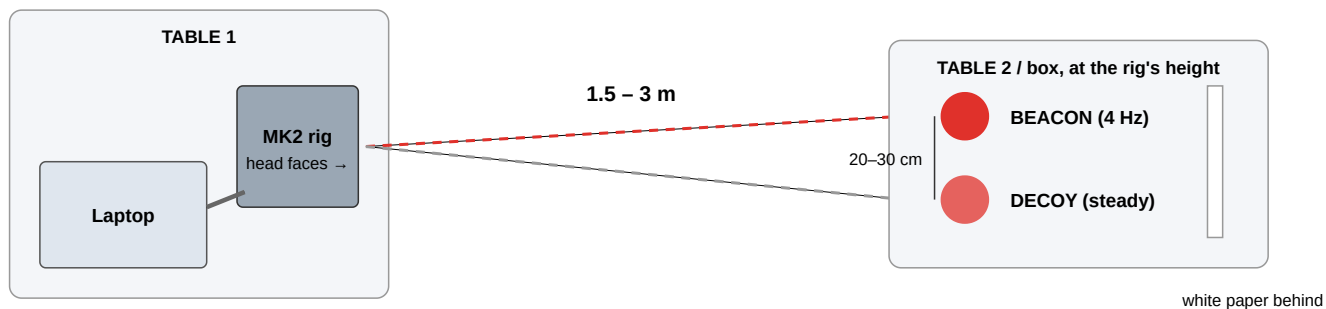
6 Beacon on, 1–3 m away.

CHECK It locks (green box). The head turns until the beacon is in the centre. The GIMBAL readout ends in **enc** (encoder confirmed).

7 Bright room: the red filter over the camera is what makes this work. If the room is still too bright, add a second layer or dim the lights. The **EXPOSURE** slider helps where the browser supports it (Windows/Android).

10.8 Demo layout

Top view of the demo table



Same distance and same height for both targets, 20–30 cm apart, both inside the camera view. A sheet of white paper behind them makes the laser dot easy to see.

10.9 Demo script (2 minutes)

- 1 Lock.** Beacon on. The rig finds it, turns until it's centred, and the laser dot lands on it (after SET AIM). The page shows LOCKED and about 4.0 Hz.
- 2 Decoy.** Switch the decoy on: same colour, same size, just as bright. The lock stays on the beacon. *"Brightness is not identity. The blink is."*
- 3 Follow.** Pick the beacon up and move it slowly left, right, up and down. The rig follows; the decoy is ignored.
- 4 Wrong rate.** On the page, set BEACON RATE to **6 Hz**. The rig drops the 4 Hz beacon, because it only locks the rate it was told to. Set it back to 4 Hz and it locks again.
- 5 Lost and found.** Cover the beacon with your hand for 3 seconds. The rig holds, then searches. Uncover it and it locks again.

11 Calibrate (15 minutes)

- 1 Straight ahead = zero.** Kill switch OFF, point the head straight ahead by hand, then press **CONNECT RIG**. Connecting restarts the Nano, and it takes the current position as zero. Then switch the motors ON. (In the Serial Monitor, **Z** does the same.)
- 2 45° check** (Serial Monitor, before connecting the website). Send **P400 T0** (45°) and check it with a printed protractor under the platform.
CHECK 45° ± 1°, and **?** reads ≈ 400. If it's off by exactly 2×, 4× or ½, the MS1-3 jumpers are wrong.

- 3 SET AIM** (the important one). Put a sheet of white paper behind the beacon. When the beacon is locked, the laser switches on by itself (blinking at 7 Hz). Click **SET AIM**, then click exactly on the laser dot in the video.

CHECK From now on the rig puts the beacon on the laser dot, not the image centre. This cancels the camera–laser offset (parallax).

- 4 Balance.** Motors off, leave the head at +15° and at –15°. If it drops, move the camera/laser/standoff until it stays. A balanced head doesn't sag or lose steps.

12 Troubleshooting

SYMPTOM	MOST LIKELY CAUSE	FIX
Motor doesn't move, no holding	RST–SLP jumper missing; no 12 V; EN not to GND	Wires 13/14; switch on; meter VMOT ≈ 12 V
Motor buzzes / shakes, doesn't turn	Coil pairs mixed (1A with 2A)	Re-find pairs with the meter; one pair on 1A/1B, the other on 2A/2B
Moves but angle is 2×/4×/½ off	MS1-3 not all at 5 V	Wire 12 (all three HIGH = 1/16)
Misses steps / loses position	Vref too low, unbalanced head, cable pulling	Vref +10 %, balance the head (11.4), free the service loop
Driver very hot / shuts off	Vref too high	Lower Vref; stick a small heatsink on the chip
<code>encoders=no</code>	SDA/SCL swapped; TCA not powered; A0-A2 floating; headers not soldered	A4 → SDA, A5 → SCL; 5 V on VIN; A0-A2 to GND; solder headers
? encoder number doesn't change	Axial magnet, or chip off-axis / gap too big	Magnet test (4.4); centre chip; gap 1–2 mm
Encoder sign opposite to command	AS5600 DIR	Move DIR from GND to 5 V (8.5)
Upload fails "not in sync"	Clone bootloader	Processor → ATmega328P (Old Bootloader)
CONNECT RIG: port busy / nothing	Serial Monitor still open; Safari	Close Serial Monitor; use Chrome/Edge
MK2 CAMERA shows the laptop camera	Webcam not plugged / needs permission	Plug it in, reload, allow camera, press MK2 CAMERA (USB) again
Won't lock in a bright room	Camera over-exposed	Red filter over the lens (2 layers if needed), dim lights
Locks, then turns away from the beacon	Axis direction	INVERT / SWAP in the RIG card, TEST MOTION again
Beacon on the centre, not on the laser dot	Aim not set	SET AIM (11.3)
Beacon stops after ~30 s	Power bank switches off at low current	Use the laptop USB, another power bank, or the 5 V adapter (9A.6)
Decoy sometimes gets locked	A loose wire makes it flicker, which looks like a blink	Solder or retape the decoy joints; fresh cells
Beacon LED never lights	LED backwards, or wired to a pin other than D9	Swap the LED legs; check D9 and GND (9A.3)

Wrong rate shown / not locking	BEACON RATE $\neq$ beacon	Set BEACON RATE to the beacon's F value (4 Hz default)
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Quick reference

NANO PIN	GOES TO	NANO PIN	GOES TO
D2 / D3	PAN STEP / DIR	A4 / A5	TCA9548A SDA / SCL
D4 / D5	TILT STEP / DIR	5V	VDD $\times 2$, MS1-3 $\times 2$, TCA VIN, AS5600 VCC $\times 2$
D7	Laser S	GND	Common ground (battery $-$, drivers, TCA, AS5600, laser, EN, A0-A2, DIR)

ZeroDrift · Team ZeroDrift, Jabalpur Engineering College · SIH 2026 PS26169. Firmware: tools/rig/rig_firmware_v2.ino. Geometry: docs/cad (same source as the 3D assembly).